FEDF PROJECT

Section-12
2520030471- Ananya Roy
2520030447- M Sri Nithya

MEDIVAULT

Purpose of the Project

The purpose of the Medical Records Management System is to develop a comprehensive digital platform that enables users to securely store, manage, and access their personal healthcare information from a single centralized location. The system is designed to eliminate the difficulties of maintaining physical medical records and reduce the risk of losing important healthcare documents.

The application allows users to organize medical documents such as prescriptions, laboratory reports, scan results, vaccination records, and medical bills in a structured manner, making them easily accessible whenever required. It also provides appointment management features that help users schedule, track, and monitor consultations with doctors and healthcare providers.

In addition, the system maintains a complete medical history timeline, allowing users to view and track their healthcare journey over time. Notification and reminder features ensure that users do not miss important appointments, follow-ups, or health-related activities. The platform also stores essential emergency medical information, including blood group, allergies, existing medical conditions, and emergency contacts, enabling quick access during critical situations.

Overall, the project aims to improve the organization, accessibility, security, and management of personal healthcare records while providing a convenient, efficient, and user-friendly solution for individuals to maintain their medical information digitally.

Problem Statement

Managing personal medical records through physical documents is often difficult, time-consuming, and prone to loss or damage. Medical reports, prescriptions, bills, vaccination records, and appointment details are usually scattered across multiple sources, making them difficult to access when needed. Additionally, the lack of a centralized system can lead to poor record organization, missed appointments, and difficulty in tracking medical history. Therefore, there is a need for a secure and organized digital solution that enables users to store, manage, and access their healthcare information efficiently from a single platform.

Goal of the Project

The goal of the Medical Records Management System is to provide a centralized and secure platform for storing, organizing, and managing personal healthcare information. The system aims to improve accessibility to medical records, simplify appointment tracking, maintain a complete medical history, and help users efficiently manage their healthcare data through a user-friendly digital solution.

Scope

In Scope

- User registration and login.
- Storage and management of medical records such as prescriptions, lab reports, scan reports, vaccination records, and medical bills.
- Appointment scheduling and tracking.
- Medical history timeline for viewing records and appointments chronologically.
- Notifications and reminders for upcoming appointments.
- Management of emergency medical information (blood group, allergies, medical conditions, emergency contacts).
- Responsive user interface with light and dark mode support.
- Local data storage with persistence across sessions.

Out of Scope

- Direct integration with hospitals, clinics, or healthcare providers.
- Online consultation or telemedicine services.

- Real-time health monitoring through wearable devices.
- Electronic prescription generation by doctors.
- Online payment processing for medical services.
- AI-based disease diagnosis or medical recommendations.
- Multi-user record sharing and access control between patients and healthcare professionals.
- Cloud-based storage and synchronization across devices.

Users

1. **Patients / Individuals**
    - Store and manage personal medical records.
    - Track appointments and medical history.
    - Maintain emergency medical information.
2. **Family Members / Caregivers** *(if authorized by the user)*
    - Help manage and monitor medical records and appointments.
    - Access emergency information when required.
3. **System Administrator**
    - Manage system functionality and maintain the application.
    - Monitor and update system settings if applicable.

Functional Requirements

User Management

- Users should be able to register a new account.
- Users should be able to log in and log out.
- The system should maintain user sessions using local storage.
- Users should be able to update their profile information.

Medical Records Management

- Users should be able to upload medical documents.
- Users should be able to view uploaded records.
- Users should be able to download records.
- Users should be able to delete records.
- The system should categorize records (Prescriptions, Reports, Bills, Scans, etc.).
- Users should be able to search and filter records.

Appointment Management

- Users should be able to add medical appointments.
- Users should be able to edit appointment details.
- Users should be able to delete appointments.
- Users should be able to mark appointments as completed or cancelled.
- Users should be able to view upcoming appointments.

Medical History Timeline

- The system should display medical records and appointments in chronological order.
- Users should be able to review their medical history through a timeline view.

Notifications

- The system should generate appointment reminders.
- The system should notify users about important updates and activities.
- Users should be able to mark notifications as read.

Emergency Information

- Users should be able to store emergency contact details.
- Users should be able to store blood group, allergies, and medical conditions.
- The system should display emergency information when required.

Theme Management

- Users should be able to switch between light mode and dark mode.
- The selected theme should be saved and restored automatically.

Data Persistence

- The system should store data using localStorage.
- User data should remain available after page refresh or browser restart.

Non-Functional Requirements

Performance

- The application should load quickly.
- Page navigation should be smooth and responsive.
- Data retrieval from localStorage should occur without noticeable delay.

Reliability

- User data should not be lost during refresh.
- The system should consistently restore stored information.
- The application should handle invalid inputs gracefully.

Security

- User authentication should restrict access to protected pages.
- Sensitive user information should not be exposed unnecessarily.
- Session management should prevent unauthorized access.

Usability

- The interface should be easy to understand and use.
- Forms should provide clear validation messages.
- Navigation should be intuitive for users of all ages.

Maintainability

- The code should be modular and reusable.
- Components should be organized for future enhancements.
- The application should support easy debugging and updates.

Scalability

- The system architecture should allow future backend integration.
- New modules and features should be easy to add.

Compatibility

- The application should work on major web browsers.
- The application should be responsive on desktops, tablets, and mobile devices.

Accessibility

- The interface should provide readable text and proper contrast.
- The application should support keyboard navigation where possible.
- Dark mode should improve accessibility in low-light environments.

Availability

- The system should be accessible whenever the browser is available.
- Stored data should remain available across sessions.

User Experience

- The application should provide a professional healthcare-themed design.
- Dark mode and light mode should offer a consistent experience.
- The overall design should feel modern, clean, and responsive.

Assumptions

- Users have access to a modern web browser.
- Users provide accurate medical information.
- Supported file formats are PDF and image files.
- Data is stored using localStorage.
- Browser storage is not manually cleared by users.

- The system is intended for personal medical record management.
- Internet connection is available for accessing the application.
- Future hospital and cloud integrations are outside the current scope.

Acceptance Criteria

User Authentication

- Users can successfully register a new account.
- Users can log in using valid credentials.
- Users remain logged in after page refresh.
- Users can log out successfully.

Medical Records Management

- Users can upload medical documents.
- Uploaded records are displayed correctly.
- Users can view, download, and delete records.
- Uploaded records persist after page refresh.

Appointment Management

- Users can create appointments.
- Users can edit appointment details.
- Users can delete appointments.
- Appointment information remains available after refresh.

Medical History Timeline

- Records and appointments are displayed in chronological order.
- Timeline updates automatically when new data is added.

Notifications

- Notifications are generated for relevant activities.
- Users can view and manage notifications.

Profile Management

- Users can update profile information.
- Changes are saved and displayed correctly.

Emergency Information

- Users can add emergency details.
- Emergency information is displayed correctly when accessed.

Dark Mode

- Users can switch between light and dark themes.
- Theme preference remains saved after refresh.

Data Persistence

- All application data is stored successfully.
- Data remains available after browser refresh or reopening the application.

User Interface

- The application is responsive on desktop and mobile devices.
- Navigation between pages works correctly.
- Forms provide appropriate validation messages.

Overall System

- The application performs without major errors.
- All modules function as intended.
- The system provides a centralized platform for managing medical records and appointments.